LOGICAL DESCRIPTION

The CPE provides the arithmetic, logic and register functions of a 2-bit wide slice through a microprogrammed central processor. Data from external sources such as main memory, is brought into the CPE on one of the three separate input busses. Data being sent out of the CPE to external devices is carried on either of the two output busses. Within the CPE, data is stored in one of eleven scratchpad registers or in the accumulator. Data from the input busses, the registers, or the accumulator is available to the arithmetic/logic section (ALS) under the control of two internal multiplexers. Additional inputs and outputs are included for carry propagation, shifting, and micro-function selection. The complete logical organization of the CPE is shown below.

MICRO-FUNCTION BUS AND DECODER

The seven micro-function bus input lines of the CPE, designated F_0-F_6, are decoded internally to select the ALS function, generate the scratchpad address, and control the A and B multiplexers.

M-BUS AND I-BUS INPUTS

The M-bus inputs are arranged to bring data from an external main memory into the CPE. Data on the M-bus is multiplexed internally for input to the ALS.

The I-bus inputs are arranged to bring data from an external I/O system into the CPE. Data on the I-bus is also multiplexed internally, although independently of the M-bus, for input to the ALS Separation of the two busses permits a relatively lightly loaded memory bus even though a large number of I/O devices are connected to the I-bus. Alternatively, the I-bus may be wired to perform a multiple bit shift (e.g., a byte exchange) by connecting it to one of the output busses. In this case, I/O device data is gated externally onto the M-bus.

SCRATCHPAD

The scratchpad contains eleven registers designated R_0 through R_9 and T. The output of the scratchpad is multiplexed internally for input to ALS. The ALS output is returned for input into the scratchpad.

ACCUMULATOR AND D-BUS

An independent register called the accumulator (AC) is available for storing the result of an ALS operation. The output of the accumulator is multiplexed internally for input back to the ALS and is also available via a three-state output buffer on the D-bus outputs. Conventional usage of the D-bus is for data being sent to the external main memory or to external I/O devices.

A AND B MULTIPLEXERS

The A and B multiplexers select the two inputs to the ALS specified on the micro-function bus. Inputs to the A-multiplexer include the M-bus, the scratchpad, and the accumulator. The B-multiplexer selects either the I-bus, the accumulator, or the K-bus. The selected B-multiplexer input is always logically ANDed with the data on the K-bus (see below) to provide a flexible masking and bit testing capability.

ALS AND K-BUS

The ALS is capable of a variety of arithmetic and logic operations, including 2's complement addition, incrementing, and decrementing, plus logical AND, inclusive-OR, exclusive-NOR, and logical complement. The result of an ALS operation may be stored in the accumulator or one of the scratchpad registers. Separate left input and right output lines, designated LI and RO, are available for use in right shift operations. Carry input and carry output lines, designated CI and CO are provided for normal ripple carry propagation. CO and RO data are brought out via two alternately enabled tri-state buffers. In addition, standard look ahead carry outputs, designated X and Y, are available for full carry look ahead across any word length.

The ability of the K-bus to mask inputs to the ALS greatly increases the versatility of the CPE. During non-arithmetic operations in which carry propagation has no meaning, the carry circuits are used to perform a word-wise inclusive-OR of the bits, masked by the K-bus, from the register or bus selected by the function decoder. Thus, the CPE provides a flexible bit testing capability. The K-bus is also used during arithmetic operations to mask portions of the field being operated upon. An additional function of the K-bus is that of supplying constants to the CPE from the microprogram.

MEMORY ADDRESS REGISTER AND A-BUS

A separate ALS output is also available to the memory address register (MAR) and to the A-bus via a three-state output buffer. Conventional usage of the MAR and A-bus is for sending addresses to an external main memory. The MAR and A-bus may also be used to select an external device when executing I/O operations.

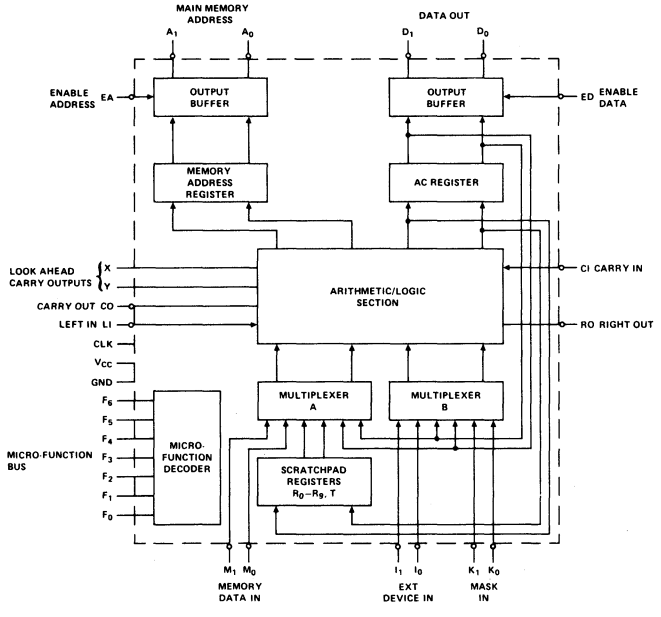


Figure 2. 3002 Block Diagram